

Lab 3: Component Modelling & Architectural Pattern Selection

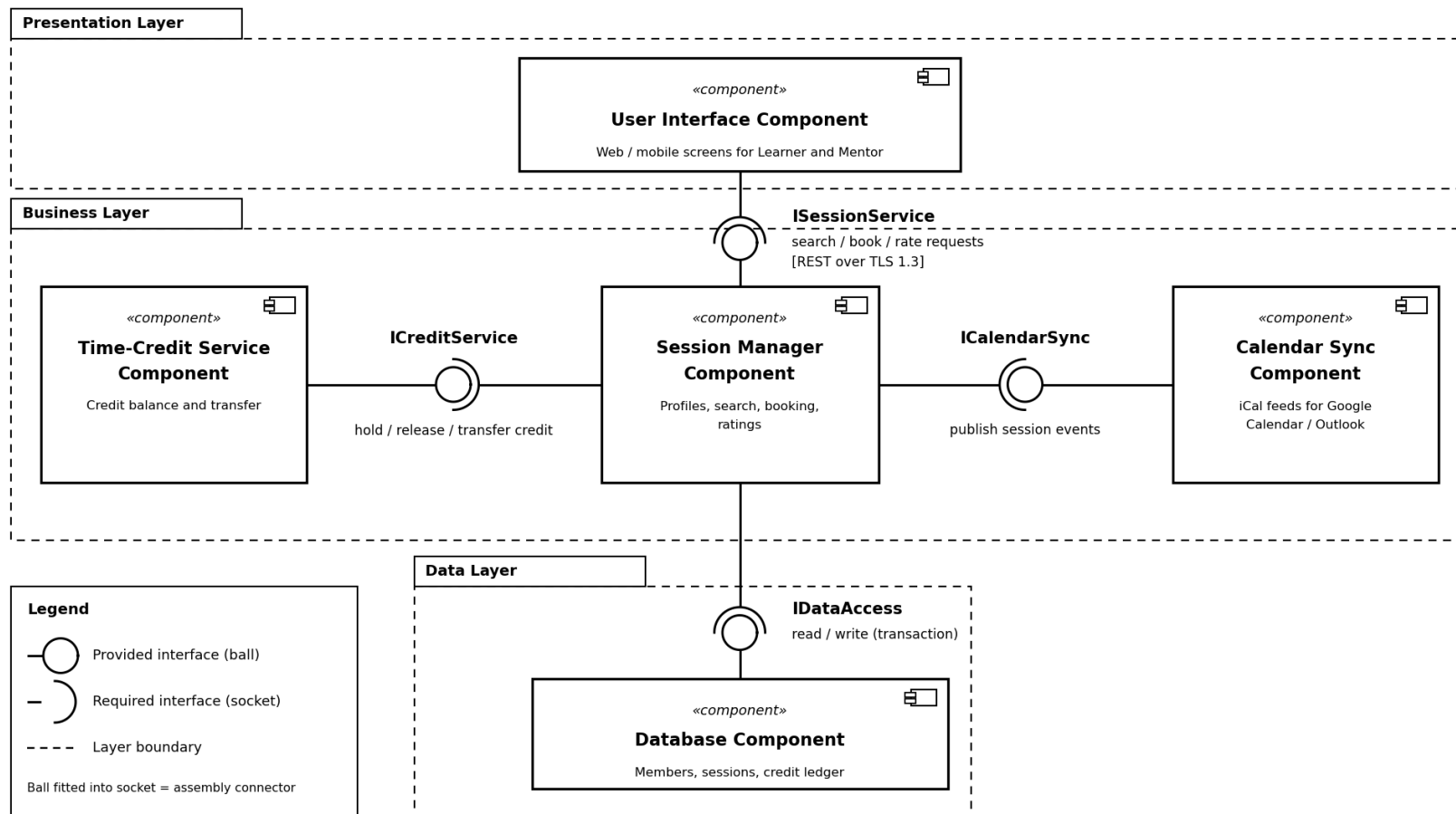
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Component Diagram

Peer Skill Exchange & Mentorship Network - Layered Architecture



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Lab 3: Architectural Justification

1. Architecture Selection

We chose Layered Architecture for the Peer Skill Exchange & Mentorship Network System. A Presentation Layer (User Interface), a Business Layer (Session Manager, Time-Credit Service, Calendar Sync) and a Data Layer (Database) are joined by four interfaces: ISessionService, ICreditService, ICalendarSync and IDataAccess.

Alternatives considered. Microservices give independent scaling and fault isolation, but their operational complexity, network latency and distributed-transaction problems (a credit moving without its booking, or the reverse) are unjustified for five cohesive components. Client-Server gives centralised control but says nothing about how server logic is structured and risks a single point of failure; a layered design can still be deployed client-server.

2. Two Reasons for the Choice

Reason 1 – Loosely coupled features and clear separation. Profiles, mentor search, booking and ratings (FR-002 to FR-005) run through the Session Manager, while credit accounting (FR-001) and calendar sync (NFR-001) sit behind their own interfaces. Because components talk only through interfaces, each feature can be built and tested on its own and plugged in (credits first, then booking, then calendar). Delivery therefore starts incremental, with every feature loosely connected, and moves to iterative refinement of each one as feedback arrives. A new calendar provider can be added behind ICalendarSync without touching the UI.

Reason 2 – Business rules, locking and rollback in one place. Rules such as blocking a booking at 0 credits and moving credit only after sign-off (FR-001) must never be bypassed. Preventing double booking (FR-004) needs the Session Manager to lock the mentor's slot the moment a learner selects it, so a second learner sees it as taken. Booking is one unit of work: if the credit hold, slot write or database commit fails, the credit hold is released and the database transaction is rolled back, so no half-finished booking or stray credit movement remains. The Business Layer is the only place these controls live, so the UI cannot skip them.

3. Security Advantage

Layering gives a controlled path to sensitive data (NFR-002), and we apply zero-trust on top of it: no component is trusted because of its position in the dependency chain. Every call on every interface is authenticated and authorised with a short-lived token (UI to Session Manager) or service credential (between Business Layer components), and nothing is assumed safe because it is “internal”. If one component is compromised, it cannot reach the ledger or database without valid, least-privilege credentials. The UI never touches the Database directly; only the Time-Credit Service changes balances. Encryption uses TLS 1.3 on the UI link and AES-256 at rest in the Data Layer.

4. Performance Benefit

A booking follows a short, predictable path: UI → Session Manager → Time-Credit Service / Database, with only three layers to cross. The UI sends independent requests (mentor search, credit balance, profile data) in parallel and non-blocking, not one after another, so screens load faster. The calendar update is published asynchronously after the booking commits, with retry on failure, so it never delays the learner's response or forces a rollback. Slot locks are held only for the short booking transaction, keeping contention low under peak load (NFR-001).

Components (5): User Interface, Session Manager, Time-Credit Service, Calendar Sync, Database. **Interfaces (4):** UI–Session Manager, Session Manager–Time-Credit Service, Session Manager–Calendar Sync, Session Manager–Database.
